

SEMESTER-IV

UMT405: SENSORS AND ACTUATORS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To provide students with an understanding of the design, analysis, and application of sensors and actuators for dynamical systems. Students will learn about different types of sensors and actuators, their operating principles, and the mechanisms used to convert physical signals into electrical signals and vice versa. Through laboratory experiments and projects, students will learn to select, design, and analyze sensors and actuators for different applications. Additionally, students will learn to integrate sensors and actuators into mechanical systems.				
Syllabus Introduction: Classification of sensors and actuators, performance characteristics: least square polynomials and data fitting, impedance and impedance matching, static and dynamic characteristics, frequency response, calibration, excitation, deadband, reliability. Application of sensors for manipulator system: shaft encoders, potentiometers, strain gauge, piezoelectric accelerometers, LVDTs. Application of sensors for autonomous mobile robots: Navigation Sensors: odometry, capacitive accelerometer, gyrometer, magnetometer, barometer, Inertial Measurement Unit, Inertial Navigation System, GPS, Introduction to complementary and Kalman filtering; Proximity Sensors: ultrasonic sensor, bump sensor; vision sensor: RGB and depth camera. Application of sensors for automation industries: Temperature Sensors, Pressure sensors, Level sensors, Flow sensors, Infrared sensors, Proximity sensors, Smoke sensors, Optical Sensors, MEMS Sensors. Actuators: Electric Linear, Electric Rotary, Fluid Power Linear, Fluid Power Rotary, Linear Chain Actuators, Manual Linear, Manual Rotary, soft actuators Interfacing with microcontroller: Interfacing methods and circuits, Case studies: Interfacing of sensors and actuators with microprocessor or microcontroller.				
Laboratory Work Measurement of Linear Displacement, Angular displacement, Temperature, Thermistor and RTD for temperature measurement. Measurement of displacement using LVDT and Capacitive transducer, Position and velocity measurement using encoders. RTDs, thermocouples, capacitive accelerometers, and gyrometer.				
Course Learning Outcomes (CLOs) The students will be able to: 1. apply different methods for measuring physical parameters 2. elucidate the functioning and application of sensors for robotics and automation 3. elucidate the working and application of the different types of actuators 4. validate the working of sensors and actuators using a computational unit				
Text Books				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

1. Sensors, Actuators, and Their Interfaces, Nathan, I., IET 2nd edition, 2020
2. Sensors and Transducers, Patronabis, D., Prentice Hall India, 2011.
3. Measurement Systems: Applications and Design, Doebelin, E.O. and Manic, D.N., 7th edition, McGraw-Hill, 2019.

Reference Books

1. Process Dynamics and Control, Seborg, D.E. and Edgar, T., John Wiley and Sons, 1989
2. Transducers and Instrumentation, Murthy, D.V.S., Prentice Hall of India, 2003.
3. Instrumentation, Measurement and Analysis, Nakra, B.C. and Chaudhry, K.K., Tata McGraw Hill, 2003.
4. Micromechanics and MEMS, Trimmer, W., IEEE Press, 1990

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
3	Sessional (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	35

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